

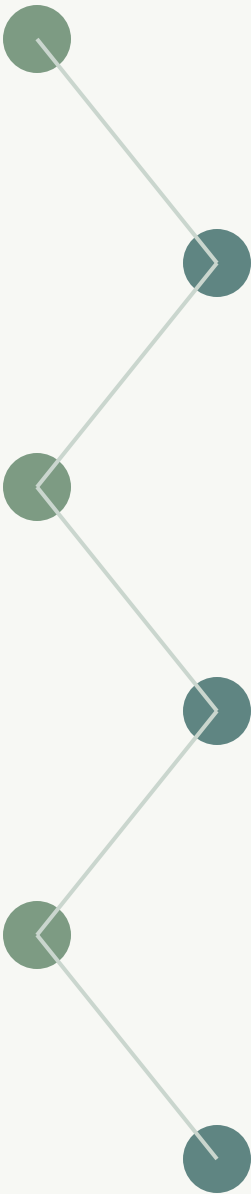
Research Knowledge Repository with Semantic Search

Literature Review & Research Gap

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Why Semantic Research Search Is Needed

02

KEYWORD SEARCH

Exact words only
Terminology mismatch
High noise
Limited context

SEMANTIC SEARCH

Understands meaning
Connects terminology
Improves relevance
Interprets context

Need: A repository that understands the meaning of research queries and retrieves relevant papers.

01

**Develop centralized research
document repository**

02

**Enable uploading &
management of papers**

03

**Implement semantic search
understanding user intent**

04

**Rank documents by
relevance**

05

**Provide user-friendly search
interface**

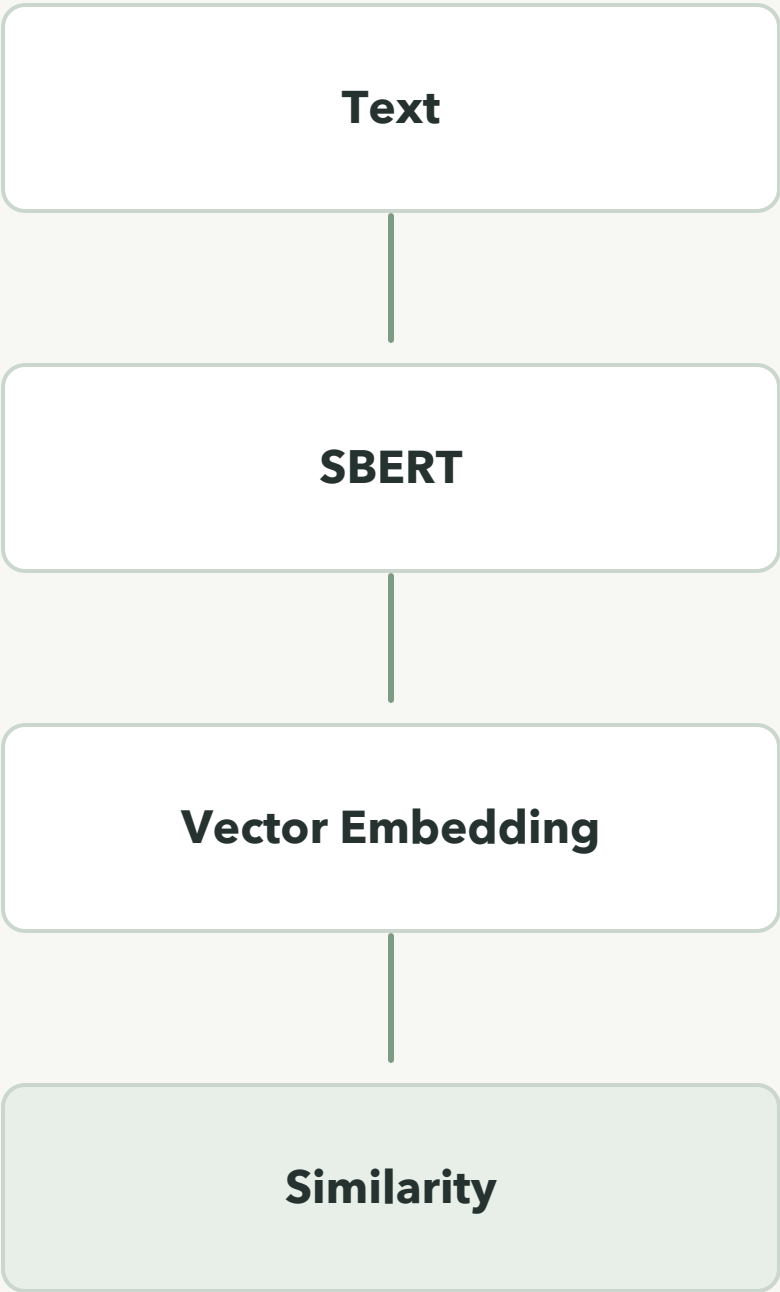


ANALYSIS LENS

Problem addressed · Dataset · Search method · Embedding technique
Ranking · Evaluation · Strengths · Limitations

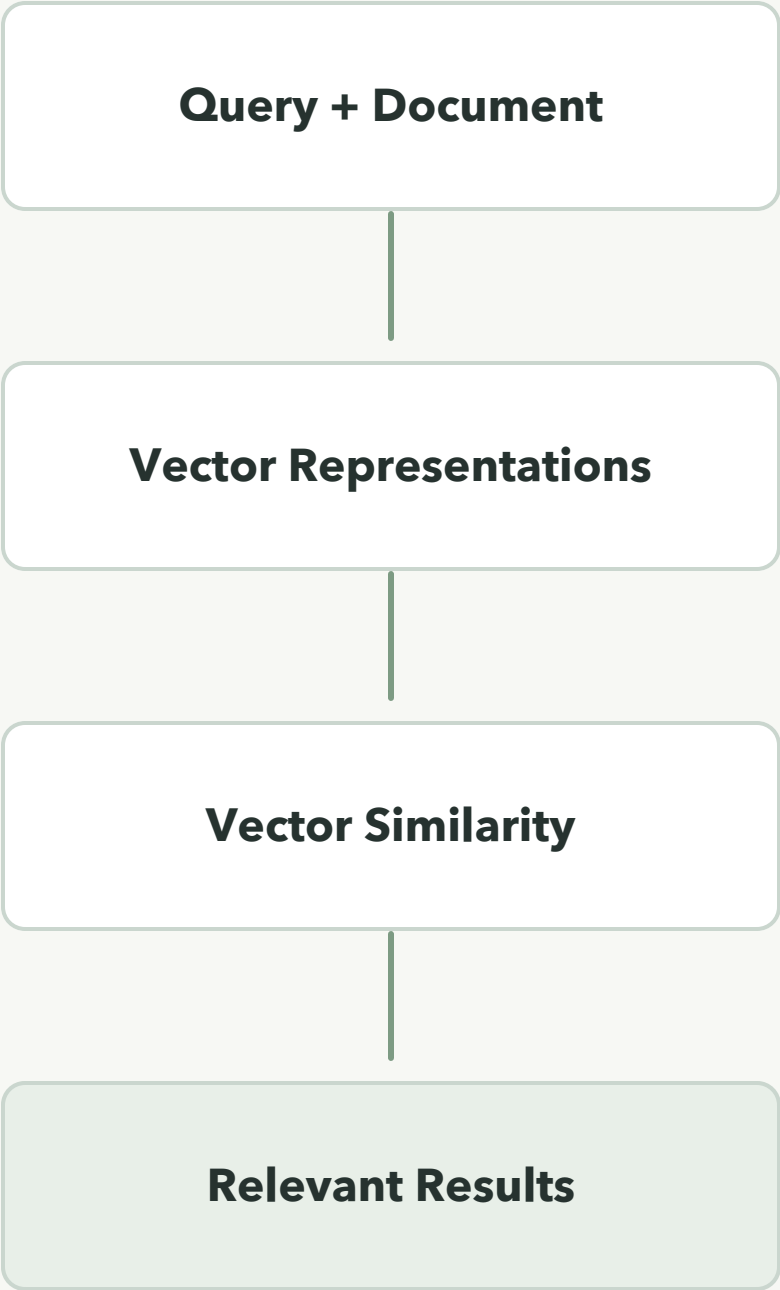
N. Reimers & I. Gurevych · 2019

PROBLEM	BERT is powerful but expensive for sentence comparison
METHOD	Siamese/triplet networks; fixed-size vectors; cosine similarity
DATASET / EVAL	Semantic textual similarity benchmarks
KEY FINDING	Efficient, meaningful sentence embeddings
STRENGTH	Efficient semantic similarity and meaning capture
LIMITATION	Embedding quality remains domain-dependent
RELEVANCE	Converts queries and papers into vectors for retrieval



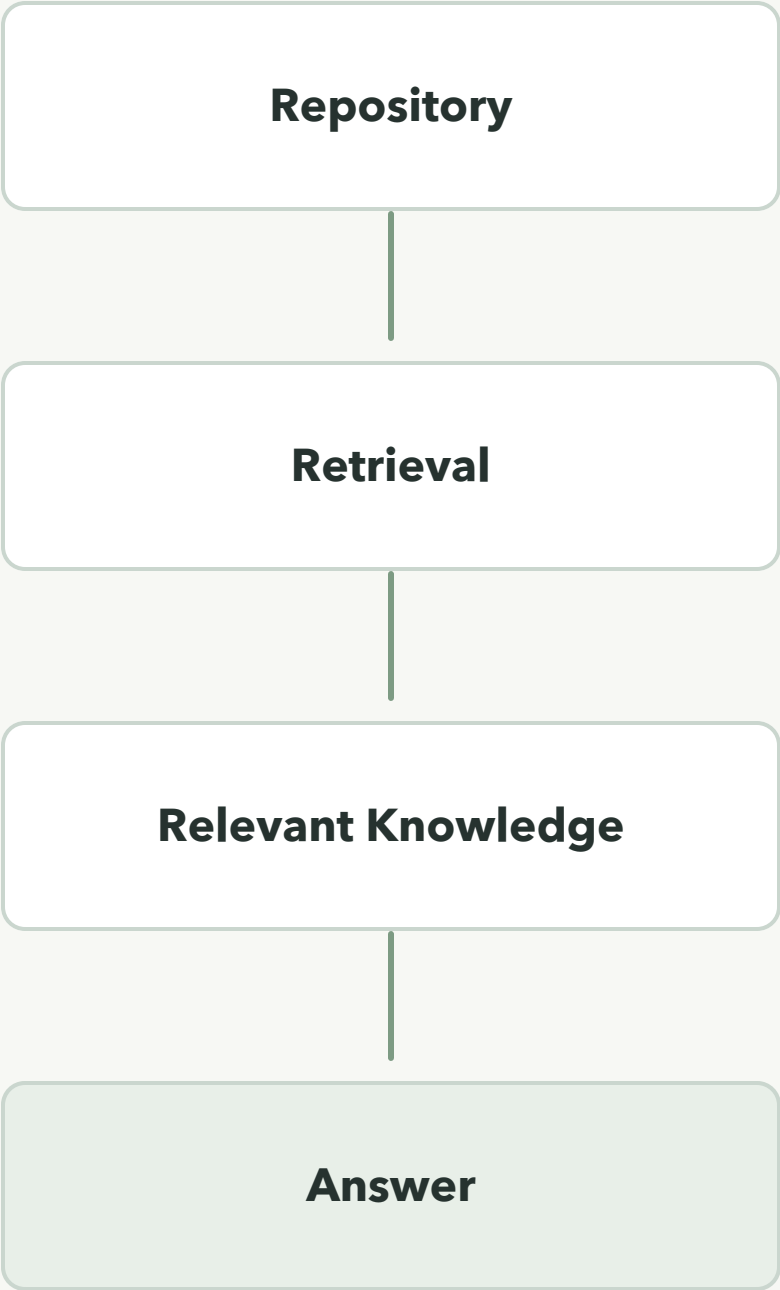
V. Karpukhin et al. · 2020

PROBLEM	Lexical matching fails with vocabulary mismatch
METHOD	Neural query/passage encoders; dense vectors; similarity retrieval
DATASET	Natural Questions and open-domain QA benchmarks
KEY FINDING	Dense retrieval outperforms sparse retrieval
STRENGTH	Meaning-based retrieval; better vocabulary handling
LIMITATION	High computation; dependent on training data
RELEVANCE	Supports semantic vector retrieval in our project



P. Lewis et al. · 2020

PROBLEM	Language models are limited by internal knowledge
METHOD	Retriever + external knowledge + generator
DATASET	NQ, WebQuestions and CuratedTREC
KEY FINDING	Retrieval improves generation performance
STRENGTH	Connects retrieval with QA and external knowledge
LIMITATION	Output depends on retrieval quality; added complexity
RELEVANCE	Future extension to QA and summarization



Paper	Main Approach	Dataset / Eval	Retrieval	Strength	Limitation
Sentence-BERT	Sentence embeddings	STS benchmarks	Vector similarity	Efficient similarity	Domain dependence
DPR	Dense neural retrieval	QA benchmarks	Dense vectors	Strong retrieval	Compute cost
RAG	Retrieval + generation	NQ / WebQ / TREC	Dense + generation	Knowledge QA	Error propagation

SBERT → DPR → RAG → OUR PROJECT

Integrated repository + semantic search + relevance ranking

COMMON FINDINGS

Semantic representations improve retrieval
Embeddings bridge vocabulary differences
Retrieval quality impacts performance
Relevance ranking is essential

DIFFERENCES

Embedding methods vary
Retrieval mechanisms differ
Datasets and evaluation vary
Ranking and final goals differ

KEY LEARNING

Integration is needed:

Document Management
+ Semantic Search
+ Ranking
+ User Access

“Existing research solves important individual components; our project integrates them into one repository.”

- 01

Dataset dependence
- 02

Limited domain coverage
- 03

Document management lacking
- 04

Retrieval without full repo functionality
- 05

Limited ranking
- 06

Limited contextual understanding
- 07

No integrated research-paper repository
- 08

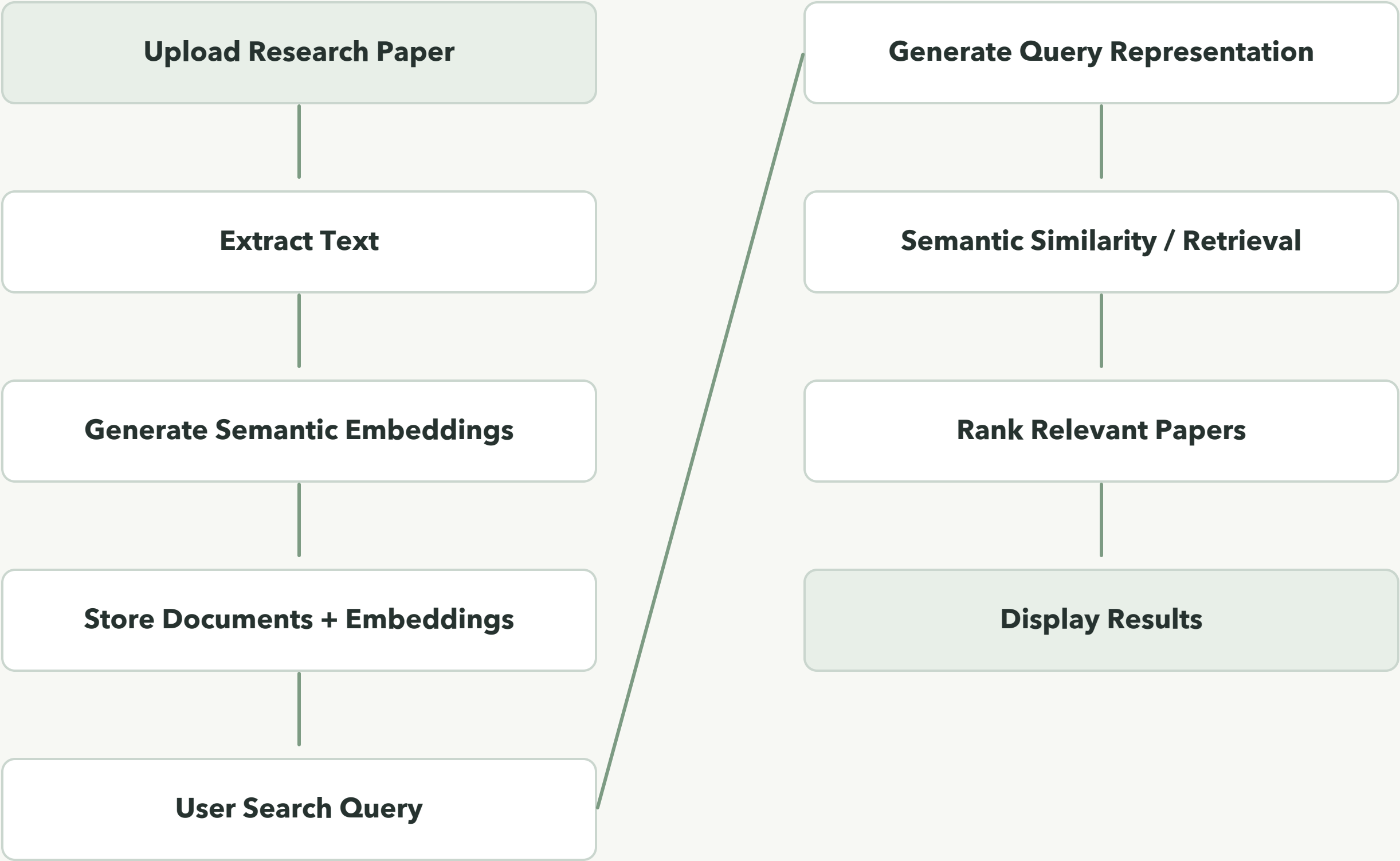
Limited summarization / QA support



Existing research addresses individual aspects of semantic retrieval, but an integrated research knowledge repository combining document management, semantic search and relevance-based retrieval is still needed.



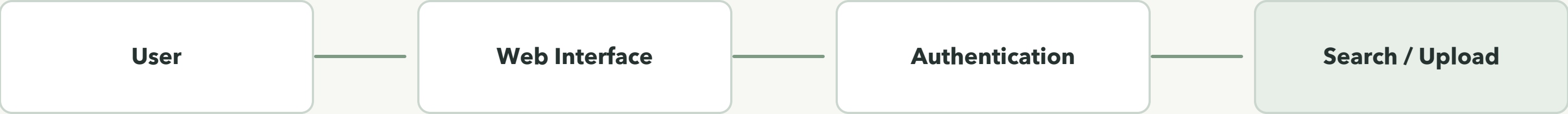
How Our Project Addresses the Gap



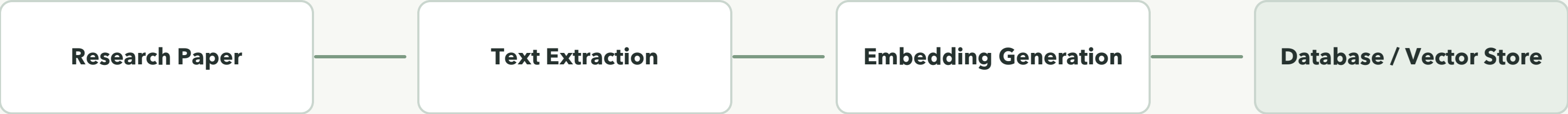
Existing Systems vs Our Proposed System

FEATURE	EXISTING / TRADITIONAL	PROPOSED SYSTEM
Search	Keyword matching	Meaning-based semantic search
Context	Limited	Context-aware
Terminology variation	Problematic	Better handling
Document repository	Limited / separate	Centralized repository
Embeddings	Not always used	Semantic embeddings
Ranking	Basic / limited	Relevance-based ranking
Future AI extensions	Limited	Summarization, QA, recommendations

USER SIDE



DOCUMENT PROCESSING



SEARCH PATH



01

**Centralized repository for
research documents**

02

**Meaning-based semantic
search**

03

**Retrieval across terminology
differences**

04

**Relevance-based document
ranking**

05

**Foundation for summarization,
QA and recommendations**



Our project aims to bridge the identified gap by combining research-document management with semantic understanding and relevance-based retrieval in a unified repository.

- [1] N. Reimers and I. Gurevych, "Sentence-BERT: Sentence Embeddings using Siamese BERT-Networks," Proceedings of EMNLP-IJCNLP, 2019.
- [2] V. Karpukhin et al., "Dense Passage Retrieval for Open-Domain Question Answering," Proceedings of EMNLP, 2020.
- [3] P. Lewis et al., "Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks," Advances in NeurIPS, 2020.

THANK YOU

Research Knowledge Repository with Semantic Search